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EDUCATION

**University of Petroleum and Energy Studies,
India**

November 2018

PhD in Geosciences

Thesis: Seismological and Gravity Field Studies
on NW Himalaya: Tectonic Implications

Berhampur University, India

April 2012

MSc. in Geophysics

Utkal University, India

March 2010

BSc. in Physics (Honors)

RESEARCH EXPERIENCE

Research Associate, Virginia Tech

January 2023-present

Department of Geosciences

- Development and successful implementation of Deep learning applications to earthquake data processing

- Passive seismic imaging of the crustal structure beneath the Hampton Roads, Southeastern Virginia
- Successful operation and management of the Hampton Roads Seismic Network
- Contributing towards writing research articles in Managed Aquifer recharge and optimizations of Injection Strategies

**Research Associate, National Geophysical
Research Institute, Hyderabad, India**

March 2021-December 2022

- Successfully carried out research towards development of earthquake resilient society in the Himalayas.
- This work resulted in a single author research article in the Journal of the Geological Society of India.
- This work was funded through National Academy of Sciences India Research Associateship.

**Research Associate, National Geophysical
Research Institute, Hyderabad, India**

October 2018-February 2021

- Delineation of partial melts and crustal heterogeneities within the crust beneath Kumaon Himalaya, India from Lg wave attenuation. This work resulted in a research article published in the Journal Scientific Reports (Nature).
- Installation, operations and maintenance of the 32-Station broadband seismic network in the high-altitude northwest Himalayan sector. This work was funded through CSIR-NGRI Research Associateship.

**Research Fellow, Wadia Institute of
Himalayan Geology, Dehradun, India**
Department of Geophysics

April 2013-June 2018

- Characterization of Seismic velocities beneath the northwestern Himalayas, India from seismic phases. This work resulted in a research article published in the Journal of Seismology (Springer).
- Seismic source characterizations and estimating the slip potential in the Western Himalayan segment, NW Himalaya, India. This work resulted in a research article published in the Journal of Asian Earth Sciences (Elsevier).
- Modeled the stress transfer mechanism and seismicity in the western Himalaya, India since 1905 and investigated their role in the incomplete ruptures of the Main Himalayan Thrust fault in the Himalayas. This work resulted in a research article in the journal Tectonics (AGU).

RESEARCH FUNDING

1. Funding Source: Hampton Roads Sanitation District – Sustainable Water Initiative for Tomorrow (SWIFT) (**Current**).

Project Title: Seismic monitoring & earthquake hazard assessment for managed aquifer recharge operations in southeast Virginia (PI Pollyea)

Objective: Monitor & mitigate induced seismicity associated with managed aquifer recharge into the Virginia Coastal Plain Aquifer.

- Duration: 60 months
- Start Date: 07/2021
- Total Value: \$1,451,052 (approx.)
- Role: Research Associate, Senior Personnel
- Brief Description of Role: Earthquake monitoring and research including real-time earthquake detection and earthquake source characterizations; Data quality control/assurance and archiving (**Antelope real-time system**); Troubleshooting, maintenance and repair of HRSN seismic stations and telemetry systems; Project reporting and client communications; Research activities including

independent research, scientific publishing, and research dissemination.

2. Funding Source: Department of Energy (**Pending**).

Project Title: Central Appalachian Regional Depot for Industrial CO₂ Abatement in Lone Star, Virginia (Project CARDINAL) (PI Pollyea)

Objective: CarbonSAFE Phase II feasibility study at Roanoke Cement Plant

- Duration: 36 months
- Start Date: 10/2024
- Total Value: \$9,000,000 (approx.)
- Role: Senior personnel

3. Funding Source: National Science Foundation (**Pending**)

Project Title: Natural and Induced Earthquake Classification using Input- and Output-Constrained Physics Informed Neural Networks (PI Pollyea; Co-PI-Mahesh Parija).

Objective: Application of Input- and Output-Constrained Physics-Informed Neural Networks (PINNs) to improve the determination of seismic moment tensors and the classification of seismic events.

- Duration: 24 months
- Start Date: 01/2026
- Total Value: \$347,858 (approx.)
- Role: Co-PI

4. Funding Source: United States Geological Survey (**Pending**)

Project Title: Developing a High-Resolution Earthquake Catalog and Imaging Active Fault Structures in the Giles County Seismic Zone Using a Temporary STRYDE Node Network (PI Sean Bemis; Co-PI-Mahesh Parija).

Objective: Identifying active faults in the Giles County seismic zone for precise seismic hazard assessments in southwest Virginia.

- Duration: 12 months
- Start Date: 01/2026
- Total Value: \$86,505 (approx.)
- Role: Co-PI

FELLOWSHIPS

- Postdoctoral Research Associate by the National Academy of Sciences, India, 2021.
- Institute Postdoctoral Research Associate at CSIR-National Geophysical Research Institute, Hyderabad, India, 2018.
- Senior Research Fellowship at the Wadia Institute of Himalayan Geology, India, 2015.
- Junior Research Fellowship at the Wadia Institute of Himalayan Geology, India, 2013.

HONORS & AWARDS

- International Travel Grant, Science and Engineering Research Board by the Department of Science and Technology, Govt. of India, 2022.
- Student Travel Grant Award by the American Geophysical Union, Washington D.C., USA, 2018.
- Student Travel Grant Award by the American Geophysical Union, Washington D.C., USA, 2018.

PUBLICATIONS (PEER REVIEWED JOURNALS)

- Ogunleye, J.B., Conley, E.W., **Parija, M.P.**, Pollyea, R.M. (2025). The role of clay layers in managing fluid pressure propagation during managed aquifer

recharge. *Groundwater for Sustainable Development*. 31, 101529. doi: <https://doi.org/10.1016/j.gsd.2025.101529>.

- Conley, E.W., Chambers, C.R., Ogunleye, J.B., Koehn, L.W., Holloway, D., Heisig-Mitchell, J., Chapman, M., **Parija, M.P.**, Pollyea, R.M. (2025). Pressure management strategies for large-scale aquifer recharge: Mitigating the potential for injection-induced earthquakes, *Journal of Hydrology*, 653-132767. doi: <https://doi.org/10.1016/j.jhydrol.2025.132767>.
- **Parija, M.P.**, Kumar, S., V H, A. (2023). Delineation of partial melts and crustal heterogeneities within the crust beneath Kumaon Himalaya, India from L_g wave attenuation, *Scientific Reports (Nature)* 13, 9922. doi: <https://doi.org/10.1038/s41598-023-36269-z>.
- **Parija, M. P.** (2022). Earthquake Swarm and Quiescence Precursory to the NE Himalaya 2011 Sikkim Earthquake (M 6.9). *Journal of Geological Society of India*. 98, 178–180. doi: <https://doi.org/10.1007/s12594-022-1955-8>.
- Biswal, S., Kumar, S. Priestley, K., Mohanty, W.K., **Parija, M. P.** (2022). Structure and Deep Earthquakes beneath the Kinnaur, NW Himalaya: Constraints from Relocated Seismicity and Moment Tensor Analysis. (2022). *Lithosphere (GSA)*, 2022:2021 (Special 6): 2834594. doi: <https://doi.org/10.2113/2022/2834594>
- **Parija, M. P.**, Kumar, S., Tiwari, V. M., Biswal, S., Biswas, A., Velliyaithu, A. (2021). Coulomb stress modeling and seismicity in the western Himalaya, India since 1905: Implications for the incomplete ruptures of the Main Himalayan Thrust. *Tectonics*. 40, 1-16. doi: <https://doi.org/10.1029/2020TC006204>
- Biswal, S., Kumar, S., Roy, S.K., Kumar, M.R., Mohanty, W.K., **Parija, M. P.**, Paul, A. (2020). Upper Mantle Anisotropy beneath the Western Segment, NW Indian Himalaya, Using Shear Wave Splitting. (2020). *Lithosphere (GSA)*. 1-9. doi: <https://doi.org/10.2113/2020/8856812>
- **Parija, M. P.**, Kumar, S., Tiwari, V. M., Purnachandra Rao, N., Kumar, N., Biswal, S., Singh, I. (2018). Microseismicity, tectonics and seismic potential in the Western Himalayan segment, NW Himalaya, India. *Journal of Asian Earth Sciences*. 159, 1-16. doi: <https://doi.org/10.1016/j.jseaes.2018.03.016>.
- Biswas, A., **Parija, M. P.**, Kumar, S. (2017). Global nonlinear optimization for the interpretation of source parameters from total gradient of gravity and

magnetic anomalies caused by thin dyke. *Annals of Geophysics*. 60(2), 1-17. doi: <https://doi.org/10.4401/ag-7129>.

- **Parija, M.P.**, Kumar, S., Biswal, S. et al. (2016). A preliminary one-dimensional crustal velocity model for Himachal Pradesh, India. *Journal of Seismology*. 20, 305–318. doi: <https://doi.org/10.1007/s10950-015-9528-6>.
- Kumar, S., Singh, P., Singh, P., Biswal, S., **Parija, M. P.** (2016). Frequency dependent attenuation characteristics of coda waves in the Northwestern Himalayan (India) region. *Journal of Asian Earth Sciences*. 117, 337-345. doi: [10.1016/j.jseaes.2015.12.023](https://doi.org/10.1016/j.jseaes.2015.12.023).
- **Parija, M.P.**, Chapman, M.C., Biswal, S., Conley, E.W., Ogunleye, J.B., Pollyea, R.M. Crustal architecture of the Hampton Roads, Virginia coastal plain based on surface wave dispersion analysis: Implications for Anthropogenic seismicity. (2026). *The Seismic Record (TSR-SSA, under revision)*.
- Conley, E.W., Chapman, M.C., **Parija, M.P.**, Ogunleye, J.B., Pollyea, R.M. Assessing Induced Earthquake Detection for Industrial-Scale Fluid Injection Operations. (2025). *Seismological Research Letters (SRL, under revision)*.

PUBLICATIONS (CONT.): BOOK CHAPTERS

- **Parija, M.P.**, Biswal, S., Biswas, A.K. Chapter 13 - Static stress change from February 6, 2017 (M 5.8) earthquake Northwestern Himalaya, India, Editor(s): Pijush Samui, Barnali Dixon, Dieu Tien Bui, *Basics of Computational Geophysics*, Elsevier, 2021, P. 213-222, ISBN 9780128205136, Doi: <https://doi.org/10.1016/B978-0-12-820513-6.00020-5>.
- Trivedi, S., Kumar, P., **Parija, M.P.**, Biswas, A. Global Optimization of Model Parameters from the 2-D Analytic Signal of Gravity and Magnetic Anomalies Over Geo-Bodies with Idealized Structure. In: Biswas, A., Sharma, S. (eds) *Advances in Modeling and Interpretation in Near Surface Geophysics*. Springer Geophysics. Springer, 2020, Cham. https://doi.org/10.1007/978-3-030-28909-6_8.

MENTORING EXPERIENCE

Mentor for Ethan W. Conley, Graduate student Fall 2024
at Virginia Tech

Current Status: Postdoctoral Fellow at the
Sandia National Lab

Mentor for John B. Ogunleye, Graduate student Fall 2024-Spring 2026
at Virginia Tech

Current Status: Joining as a Hydrogeologist at
the Kansas state Geological Survey

SERVICE & OUTREACH

- Guest Editor for the Advances in statistical and computational seismology Collection - Nature Scientific Reports, 2024-Present.
- Editorial Board Member for Nature Scientific Reports (Springer-Nature), 2023–Present.
- Editorial Board Member for Nature Scientific Reviews (Springer-Nature), 2026–Present.
- Reviewer for Journal of Geophysical Research-Solid Earth (AGU: Print ISSN:2169-9313), 2024–Present
- Reviewer for Natural Hazards (Springer Nature: Online ISSN:2169-9313, 2024–Present
- Reviewer for International Journal of Geophysics (Hindawi: Print ISSN:1687-885X), 2023-Present.
- Reviewer for Journal of Earth System Sciences (Springer Nature: Electronic ISSN 0973-774X): 2021-Present.
- Reviewer for Nature Scientific Reports (Springer Nature), 2023-Present.

PRESENTATIONS

Workshop/Conference Oral Participation

- Speaker at the Joint Scientific Assembly IAGA – IASPEI 2021 at Hyderabad, India, 2021.
- Speaker at the American Geophysical Union, Fall Meeting, Washington D.C., USA, 2018.
- Speaker at the workshop on Earthquake and Tsunami: Modeling and Observations, International Center for Theoretical Physics (ICTP), Hanoi, Vietnam, 2017.
- Speaker at the National Geo-Research Scholar's meeting, Dehradun, India, 2016.
- Speaker at the National workshop on the status of Natural Hazards in Himachal Pradesh, Dharamshala, Himachal Pradesh, India, 2014.

Invited Talks

- Invited Speaker at the Hampton Roads Sanitation District-Sustainable Water Research for Tomorrow (HRSD-SWIFT) workshop, Nansemond, Virginia, USA, 27th June 2024.
- Invited Speaker at the SWIFT Research Center, Nansemond, Virginia, USA, 06th June 2023.
- Invited Speaker at Vietnam Institute for Advanced Studies in Mathematics (VIASM), Hanoi, Vietnam December 2017.

Posters

- **Parija, M. P.**, Chapman, M. C., Biswal, S., Conley, E. W., Ogunleye, J., Pollyea, R.M. (2025). Shear Wave Velocity Structure and Moho Depth Beneath the Virginia Coastal Plain (VCP) From Fundamental-mode Rayleigh-wave Group-velocity, Abstract 134, Seismological Society of America, 14-17 April, Baltimore, Maryland.
- **Parija, M. P.**, Chapman, M. C., Pollyea, R. M., Statistical analysis of the foreshock-mainshock-aftershock sequence: a case study of 2020 Mw 5.1

Sparta, North Carolina, earthquake, vol. 2023, no. 353, Art. no. S21F-0353, 2023 (2023AGUFM.S21F0353P).

- **Parija, M.P.**, Gupta, H.K. Analysis of the 2015 Gorkha-Dolakha (Central Nepal) Foreshocks and Aftershocks Sequence Through Transients in B Values. Seismological Research Letters, 2022, 93 (2B), p. 1115–1372.
- Kumar, S., Sain, K., **Parija, M.P.** et al. Re-Appraisal of Seismicity and Seismotectonics of Himalaya, India: A Review of Historical and Current Status. Journal of the Indian Geological Congress 2021, 2(2) and 13(1&2), p.45-59.
- Biswal, S., Kumar, S., Mohanty, W., **Parija, M.P.** Imaging the lithospheric structure across NW Himalaya, India utilizing local seismic tomography. Geophysical Research Abstracts, 2019, 21, p-1.
- Kumar, S., **Parija, M. P.**, and Biswal, S., “Active Tectonics and Potential of seismic hazard in the Northwest Himalayan region, India”, vol. 2018, Art. no. S14B-08, 2018 (2018AGUFM.S14B..08K).
- **Parija, M. P.**, Kumar, S., and Tiwari, V. M., “Crustal Structure, Stress Field Evolution across NW Himalaya, India from a Combined Seismic and Gravity Study”, vol. 2018, Art. no. T22A-08, 2018 (2018AGUFM.T22A..08P).
- Kumar, S., Biswal, S., and **Parija, M. P.**, “Active Tectonics of Himalayan Faults/Thrusts System in Northern India on the basis of recent & Paleo earthquake Studies”, vol. 2016, Art. no. S44B-06, 2016 (2016AGUFM.S44B..06K).
- Kumar, S., **Parija, M. P.**, and Biswal, S., “Microseismicity, Tectonics and Seismic Potential in the Western Himalayan Segment, NW Himalaya (india) Region”, vol. 2016, Art. no. S53A-2830, 2016 (2016AGUFM.S53A2830K).